

Лабораторная работа №3

К	Система	X1	X2	X3	X4	X5
20	10-я	286	293	16	4811	21
	16-я	11E	125	10	12CB	15

$$Y = [(X1 + X2)X3 - X4]/X5$$

1 шаг: Сложение

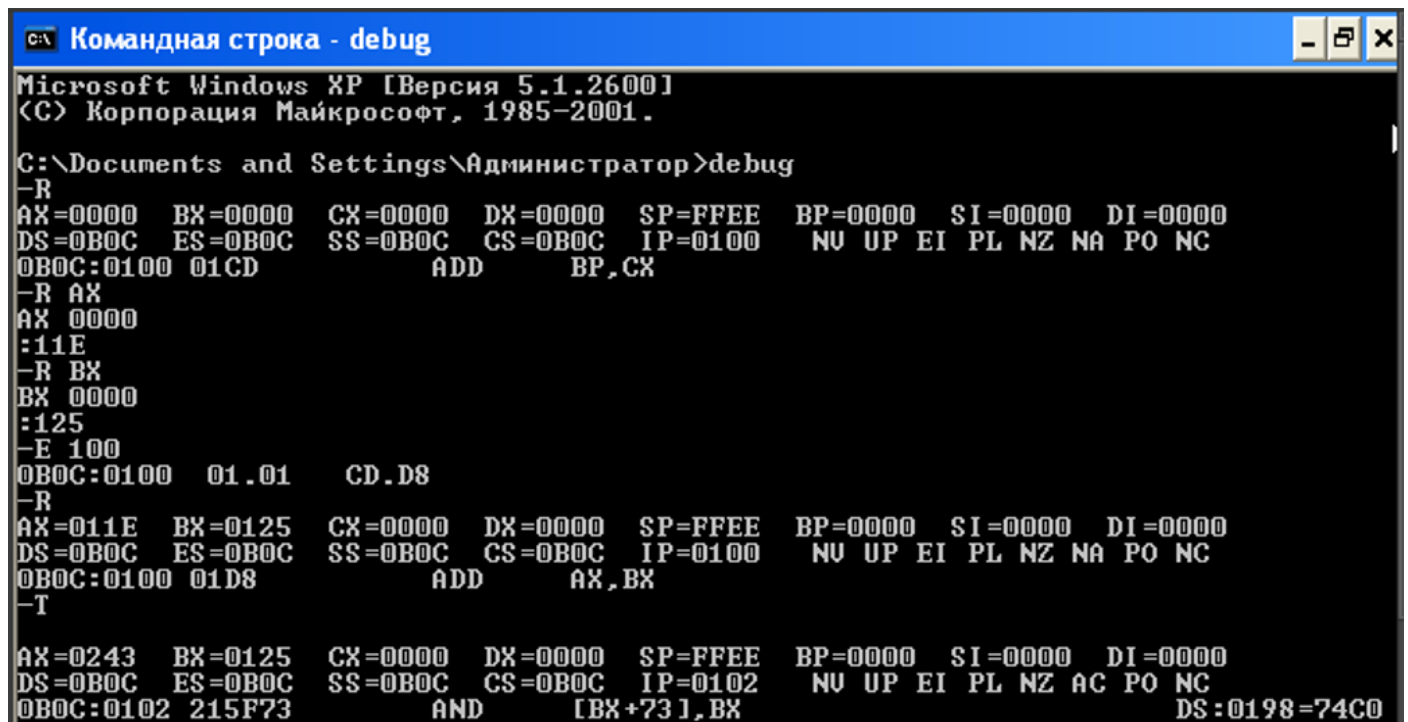
AX = записал 11E

BX = 125

0100h = D801.

вычисление с помощью ввода 'T'

Результат сложения: **243**



```
C:\Documents and Settings\Администратор>debug
-R
AX=0000 BX=0000 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100  NU UP EI PL NZ NA PO NC
0B0C:0100 01CD          ADD     BP,CX
-R AX
AX 0000
:11E
-R BX
BX 0000
:125
-E 100
0B0C:0100 01.01      CD.D8
-R
AX=011E BX=0125 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100  NU UP EI PL NZ NA PO NC
0B0C:0100 01D8          ADD     AX,BX
-T
AX=0243 BX=0125 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0102  NU UP EI PL NZ AC PO NC
0B0C:0102 215F73        AND     [BX+73],BX
DS:0198=74C0
```

2 шаг: Умножение

IP = 100

AX = 0243

BX = 10

0100h = E3F7

Результат: **2430**

```

-R IP
IP 0102
:100
-R
AX=0243 BX=0125 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100 NU UP EI PL NZ AC PO NC
0B0C:0100 01D8 ADD AX,BX
-R BX
BX 0125
:10
-E 100
0B0C:0100 01.F7 D8.E3
-R
AX=0243 BX=0010 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100 NU UP EI PL NZ AC PO NC
0B0C:0100 F7E3 MUL BX
-T
AX=2430 BX=0010 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0102 NU UP EI PL NZ NA PE NC
0B0C:0102 215F73 AND [BX+73],BX DS:0083=2020

```

3 шаг: Вычитание

IP = 100

AX = 2430

BX = 12CB

0100h = D829

Результат: **1165**

```

-R
AX=2430 BX=12CB CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100 NU UP EI PL NZ NA PE NC
0B0C:0100 29E3 SUB BX,SP
-E 100
0B0C:0100 29.29 E3.D8
-R
AX=2430 BX=12CB CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100 NU UP EI PL NZ NA PE NC
0B0C:0100 29D8 SUB AX,BX
-T
AX=1165 BX=12CB CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0102 NU UP EI PL NZ AC PE NC
0B0C:0102 215F73 AND [BX+73],BX DS:133E=1F06

```

4 шаг: Деление

IP = 100

AX = 2430

BX = 15

0100h = F3F7

DX = 1 (остаток)

Результат: **D4**

```

-R IP
IP 0102
:100
-R BX
BX 12CB
:15
-E 100
0B0C:0100 29.F7 D8.F3
-R
AX=1165 BX=0015 CX=0000 DX=0000 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0100 NU UP EI PL NZ AC PE NC
0B0C:0100 F7F3 DIV BX
-T
AX=00D4 BX=0015 CX=0000 DX=0001 SP=FFEE BP=0000 SI=0000 DI=0000
DS=0B0C ES=0B0C SS=0B0C CS=0B0C IP=0102 NU UP EI PL NZ AC PE NC
0B0C:0102 215F73 AND [BX+73],BX DS:0088=4220

```

Проверка:

$$286 + 293 = 579$$

$$579 * 16 = 9264$$

$$9264 - 4811 = 4453$$

$$\frac{4453}{21} = 212 \frac{1}{21}$$

$$212_{10} = D4_{16}$$

$$1_{10} = 11_{16}$$

$$Y = 212 \text{ и } 1 \text{ (ост.)}$$